AI2: Uncertainty in AI

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Overview

By the end of this lecture, you will be able to:

- Understand why uncertainty arises in AI systems.
- Differentiate between aleatoric and epistemic uncertainty.
- Recognize real-world applications where uncertainty is a key factor.
- Explore techniques to manage uncertainty in AI.

The Nature of Uncertainty in AI

Definition: Uncertainty refers to situations where the outcome or state of a system is not fully known.

Sources of Uncertainty:

- **Data Uncertainty:** Noisy or incomplete data.
- **Model Uncertainty:** Limited or incorrect model assumptions.
- **Environmental Uncertainty:** Dynamic, unpredictable environments.

Predictive Uncertainty

- Suppose we trained a model $f_{\hat{\mathbf{w}}}(\mathbf{x})$, where $\hat{\mathbf{w}}$ denotes the parameters of the model obtained on the training set, and $(\mathbf{x}, y)$ is a random test example.
- What is the distribution of the residual $y - f_{\hat{\mathbf{w}}}(\mathbf{x})$?
- Predictive uncertainty refers to this distribution.

Observation: Some of the predictive uncertainty can be reduced, some can't. Based on this, we talk about two main types of uncertainty.

Types of Uncertainty

Aleatoric Uncertainty:

- Intrinsic randomness in data or processes.
- Example: Sensor noise in autonomous vehicles.

Epistemic Uncertainty:

- Uncertainty due to lack of knowledge; reducible with more data or improved models.
- Example: Insufficient training data for a machine learning model.

Illustration of Uncertainties in Regression

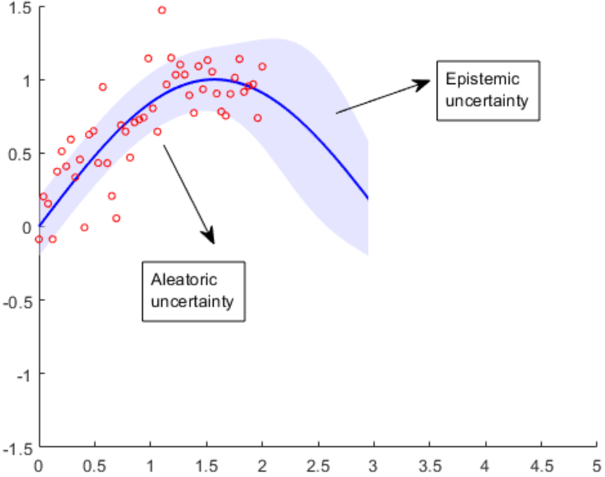

Illustration of Uncertainties in Classification

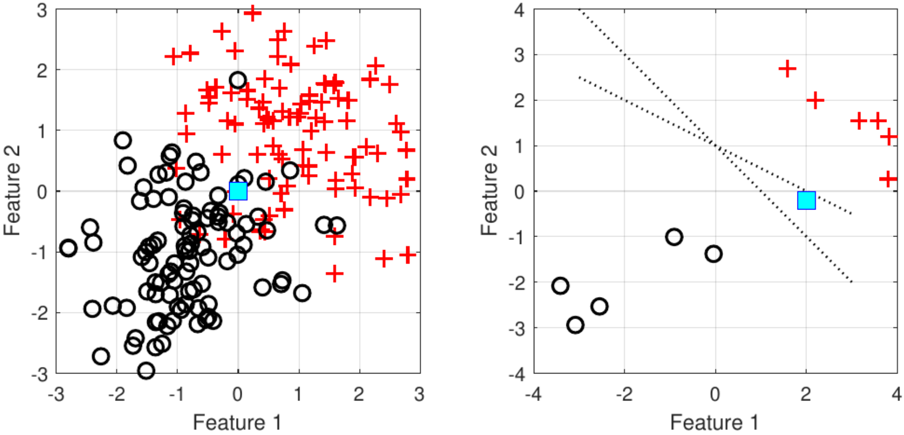

Figure: Left: Aleatoric uncertainty. Right: Epistemic uncertainty.

Aleatoric vs. Epistemic Uncertainty

- The residual error can be decomposed as:

$$y - f_{\hat{\mathbf{w}}}(\mathbf{x}) = \underbrace{y - g_{\mathbf{w}^*}(\mathbf{x})}_{\text{Aleatoric uncertainty}} + \underbrace{g_{\mathbf{w}^*}(\mathbf{x}) - f_{\hat{\mathbf{w}}(\mathbf{x})}}_{\text{Epistemic uncertainty}}$$

- **Aleatoric uncertainty**: Error of the best possible model g_{w^*}. (Typically, neither the "true model" g nor its "true parameters" $\mathbf{w}^*$ are known to us.)
- **Epistemic uncertainty**: Error of our model $f_{\hat{\mathbf{w}}}$ vs. $g_{\mathbf{w}^*}$

To simplify, let's assume that we know the model family, so $g = f$. Then,

$$y - f_{\hat{\mathbf{w}}}(\mathbf{x}) = \underbrace{y - f_{\mathbf{w}^*}(\mathbf{x})}_{\text{Aleatoric uncertainty}} + \underbrace{f_{\mathbf{w}^*}(\mathbf{x}) - f_{\hat{\mathbf{w}}(\mathbf{x})}}_{\text{Epistemic uncertainty}}$$

Note that typically $\mathbf{w}^* \neq \hat{\mathbf{w}}$ because we only see a finite amount of data.

Bayesian vs Frequentist Frameworks of handling uncertainty

Bayesian Perspective:

- Uncertainty is represented as a probability distribution (prior belief, which incorporates prior knowledge) over a set of parameter values of the model, which are updated based on new data.
- Parameters are treated as random variables, and data is seen as non-random.

Frequentist Perspective:

- Uncertainty is based on long-run frequencies of outcomes.
- Data is treated as a random draw from some (unknown) distribution, and the model parameters are non-random.
- Data is used to estimate (fit) parameters.

Key Differences Between Bayesian and Frequentist Approaches

- **Interpretation of Probability:**
  - Bayesian: Probability as a degree of belief.
  - Frequentist: Probability as the long-run frequency of events.
- **Handling of Uncertainty:**
  - Bayesian: Can quantify and update epistemic uncertainty.
  - Frequentist: Uncertainty is captured by confidence intervals or p-values, not belief.
- **Use of Data:**
  - Bayesian: Incorporates prior knowledge and continuously updates.
  - Frequentist: Relies on observed data to estimate fixed model parameters.

In this module, beyond a high-level exposition of both frameworks, we will not be concerned with purely Bayesian approaches. Hybrids making use of Bayes rule while estimating parameters is a frequentist manner tend to work well in practice.

Why handle uncertainty in AI?

- **Improved Decision-Making:** Accounts for possible errors or unknowns. Examples:
  - if uncertain, then patient should see a doctor
  - if uncertain, robot should avoid obstacle, defer action, move closer, collect more data, etc.
  - if uncertain, raise a query to the humans (active learning).
- **Better Model Generalization:** Avoids overfitting to noisy data.
- **Robustness and Reliability:**
  - Reducing epistemic uncertainty facilitates performance under unpredictable conditions.
  - Quantifying the confidence and accuracy of your model helps the human user take appropriate decisions.

Real-World Applications of Uncertainty in AI

Examples:

- **Autonomous Vehicles:** Predicting traffic behavior under varying conditions.
- **Medical Diagnostics:** Handling noisy data and ambiguous symptoms.
- **Finance:** Forecasting under market volatility.
- **Computer Vision and Robotics**
  - Many vision and robotic systems still operate with deterministic algorithms.
  - Huge scope to incorporate uncertainty handling in object detection, dynamic models, parameters, etc.
  - Your final-year-project could solve open problems!
  - You'll learn the basic building blocks in this module.

Summary

Key Points:

- Uncertainty is inherent in real-world AI systems.
- Two key types: aleatoric and epistemic uncertainty.
- Managing uncertainty improves decision-making, robustness, and reliability.

Further reading

- Hüllermeier, E., Waegeman, W. Aleatoric and epistemic uncertainty in machine learning: an introduction to concepts and methods. Machine Learning 110, 457–506 (2021).
- Rohit K. Tripathy, Ilias Bilionis: Deep UQ: Learning deep neural network surrogate models for high dimensional uncertainty quantification. Journal of Computational Physics, 2018.

A Worked Example

Epistemic and Aleatoric Uncertainty in Multivariable Linear Regression

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Linear regression

Linear regression is simple enough that we can work out explicitly the epistemic and aleatoric components of the uncertainty (and refresh our matrix-vector calculus on our way).

- Recall, the linear model (set of all linear functions) has the form

$$f_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1 x_1 + w_2 x_2, \ldots, w_d x_d$$

  where $\mathbf{w} = (w_0, w_1, \ldots, w_d)$ are parameters.
- Loss function: Mean Square Error (n training examples)

$$L(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^{n} (y_i - f_{\mathbf{w}}(\mathbf{x}_i))^2$$

- The notation above has subsumed the bias term (or intercept) by concatenating a feature of 1, i.e. our $\mathbf{x}_i$ is in fact $(\mathbf{x}_i, 1)$, for each training point.

Illustration

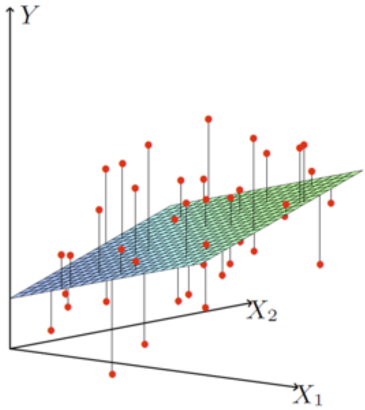


Figure: Linear regression with $d = 2$ dimensional inputs[1]

[1]From: Hastie, Tibshirani, Friedman, "The Elements of Statistical Learning", Springer, 2009.

Vectorising Linear Regression

Suppose we have n points in our training set $\{(\mathbf{x}_1, y_1), \ldots (\mathbf{x}_n, y_n)\}$, where each $\mathbf{x}_i \in \mathbb{R}^d$ is represented as a column-vector (by convention).

$$\underbrace{\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}}_{\mathbf{y}} \approx \begin{bmatrix} f_{\mathbf{w}}(\mathbf{x}_1) \\ \vdots \\ f_{\mathbf{w}}(\mathbf{x}_n) \end{bmatrix} = \begin{bmatrix} \mathbf{w}^\top \mathbf{x}_1 \\ \vdots \\ \mathbf{w}^\top \mathbf{x}_n \end{bmatrix} = \begin{bmatrix} \sum_{j=0}^d w_j x_{1,j} \\ \vdots \\ \sum_{j=0}^d w_j x_{n,j} \end{bmatrix} = \underbrace{\begin{bmatrix} x_{1,0} & \cdots & x_{1,d} \\ \vdots & \ddots & \vdots \\ x_{n,0} & \cdots & x_{n,d} \end{bmatrix}}_{\mathbf{X}} \underbrace{\begin{bmatrix} w_0 \\ \vdots \\ w_d \end{bmatrix}}_{\mathbf{w}}$$

$$\mathbf{y} \approx \mathbf{X}\mathbf{w}$$

where $\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$; $\quad \mathbf{X} = \begin{bmatrix} x_{1,0} & x_{1,1} & \cdots & x_{1,d} \\ \vdots & & \ddots & \vdots \\ x_{n,0} & x_{n,1} & \cdots & x_{n,d} \end{bmatrix}$; $\quad \mathbf{w} = \begin{bmatrix} w_0 \\ \vdots \\ w_d \end{bmatrix}$

and $x_{1,0} = \cdots = x_{n,0} = 1$.

Vectorising Mean Squared Error

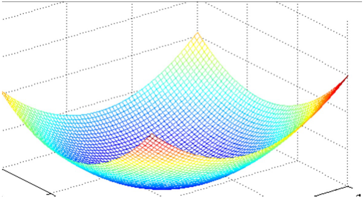
Mean Squared Error:

$$L(\mathbf{w}) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \mathbf{w}^\top \mathbf{x}_i)^2$$

Vectorized Form:

$$L(\mathbf{w}) = \frac{1}{n}\|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2$$

where $\|\mathbf{z}\|^2 = z_1^2 + z_2^2 + \dots z_d^2$ is the squared Euclidean norm of the vector in its argument.



Closed form solution

We find $\hat{\mathbf{w}}$ by minimising the MSE.

- We could use gradient descent (iterative, general applicability)
- We can analytically find the closed form solution (in the particular case of linear least square regression)

Recall, we need to minimise the loss

$$L(\mathbf{w}) = \frac{1}{n}\|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2$$

Minimum solution has gradient equal to zero

$$\nabla_{\mathbf{w}} L(\hat{\mathbf{w}}) = 0$$

First, we calculate the components of the gradient...

Closed Form Solution

- The gradient is (check for yourself):

$$\nabla_{\mathbf{w}} L(\mathbf{w}) = \nabla_{\mathbf{w}} \left[\frac{1}{n} \|\mathbf{y} - X\mathbf{w}\|^2 \right] = \nabla_{\mathbf{w}} \frac{1}{n} (\mathbf{y} - \mathbf{X}\mathbf{w})^\top (\mathbf{y} - \mathbf{X}\mathbf{w})$$

$$= \frac{1}{n} \nabla_{\mathbf{w}} \left(\mathbf{y}^\top \mathbf{y} - 2\mathbf{w}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{w}^\top \mathbf{X}^\top \mathbf{X}\mathbf{w} \right)$$

$$= -\frac{2}{n} \mathbf{X}^\top \mathbf{y} + \frac{2}{n} \mathbf{X}^\top \mathbf{X}\mathbf{w}$$

- We know that, at the minimum solution (denote it $\hat{\mathbf{w}}$), the gradient is zero, that is:

$$-\frac{2}{n} \mathbf{X}^\top \mathbf{y} + \frac{2}{n} \mathbf{X}^\top \mathbf{X}\hat{\mathbf{w}} = 0$$

- Solving for $\hat{\mathbf{w}}$:

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Assume the true model is linear

- Assume that the data is actually generated by some linear model:

$$y_i = \mathbf{w}^{*\top}\mathbf{x}_i + \epsilon_i$$

where ϵ_i is random noise independent of all other variables, and has 0-mean and variance $\mathrm{Var}(\epsilon_i) = \sigma^2, \forall i \in [d]$.

- In vectorized form:

$$\mathbf{y} = \mathbf{X}\mathbf{w}^* + \mathbf{e}, \quad \text{where } \mathbf{e} = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

- Then, we have:

$$\begin{aligned} \hat{\mathbf{w}} - \mathbf{w}^* &= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y} - \mathbf{w}^* = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top(\mathbf{X}\mathbf{w}^* + \mathbf{e}) - \mathbf{w}^* \\ &\qquad \text{(plugged in } \mathbf{y}) \\ &= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}\mathbf{w}^* + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{e} - \mathbf{w}^* \\ &= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{e} \text{ (since } (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X} = \mathbf{I}\text{, and } \mathbf{w}^* - \mathbf{w}^* = \mathbf{0}) \end{aligned}$$

Aleatoric & Epistemic Uncertainty in Linear Regression

- Let $(\mathbf{x}, y)$ be a query point with its true target. The residual error of the linear regression model's prediction decomposes as:

$$y - \hat{\mathbf{w}}^\top \mathbf{x} = \underbrace{y - \mathbf{w}^{*\top} \mathbf{x}}_{\text{Aleatoric uncertainty}} + \underbrace{\mathbf{w}^{*\top} \mathbf{x} - \hat{\mathbf{w}}^\top \mathbf{x}}_{\text{Epistemic uncertainty}}$$

- **Aleatoric uncertainty**: $y - \mathbf{w}^{*\top} \mathbf{x} = \epsilon$ (independent of n)
- **Epistemic uncertainty**: $(\mathbf{w}^* - \hat{\mathbf{w}})^\top \mathbf{x} = -\mathbf{e}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}$

Recall the noise was assumed independent with $\mathbb{E}[\epsilon_i] = 0, \mathrm{Var}(\epsilon_i) = \sigma^2$ for all $i \in \{1, 2, \ldots, n\}$.

- What is the mean (expectation) of the epistemic uncertainty?
- What is the variance of the epistemic uncertainty?

Mean of Epistemic Uncertainty

- Note that:

$$\mathbf{e}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{x} = \mathbf{x}^\top (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{e} = \mathbf{x}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \sum_{i=1}^{n} \mathbf{x}_i \epsilon_i$$

- So the mean is:

$$\mathbb{E}\left[-\mathbf{x}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \sum_{i=1}^{n} \mathbf{x}_i \epsilon_i\right] = -\mathbf{x}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \sum_{i=1}^{n} \mathbf{x}_i \mathbb{E}[\epsilon_i] = 0$$

Variance of Epistemic Uncertainty

- We have:

$$\begin{aligned}
\mathrm{Var}\left(\mathbf{x}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\sum_{i=1}^{n}\mathbf{x}_i\epsilon_i\right) &= \mathbf{x}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\sum_{i=1}^{n}\mathbf{x}_i\mathrm{Var}(\epsilon_i)\mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x} \\
&= \sigma^2\mathbf{x}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x} \\
&= \sigma^2\mathbf{x}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x} \\
&\approx \frac{\sigma^2\mathbf{x}^\top\Sigma^{-1}\mathbf{x}}{n} \quad \text{(of this order w.r.t. } n\text{)}
\end{aligned}$$

where σ^2 is the variance of the noise, and $\Sigma = \mathbb{E}[\mathbf{X}^T\mathbf{X}]$ the true covariance of the data generating distribution.

  - The variance of the epistemic uncertainty shrinks as n increases.
  - The variance of the aleatoric uncertainty stays constant σ^2.

Summary

- Under the assumption that the data generating model has the same form as the model we estimate from data, we worked out analytically the form of aleatoric and epistemic uncertainties in the case of linear least square regression.
- We should also note that, beyond this simple model, analytic calculations become more difficult, but there are statistical methods to estimate model variability that can be related to epistemic uncertainty empirically.
- Related readings:
  Hastie, T., Tibshirani, R., & Friedman, J. (2009). The Elements of Statistical Learning: Data Mining, Inference, and Prediction. Springer.
  This book covers statistical methods like the bootstrap, and while it doesn't specifically frame uncertainty as epistemic, it provides insights into how the method can capture model variability.